

Configuring Hostname and Hostname Resolution in Ubuntu Linux



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Overview

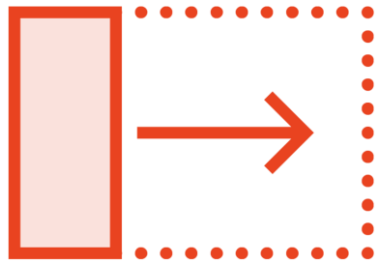


Hostnames and Resolution

- hostnamectl
 - /etc/hostname
 - /etc/machine-id
 - /etc/machine-info
 - /etc/os-release
 - systemd-hostnamed
- systemd-resolved
 - /etc/systemd/resolved.conf
 - /etc/nsswitch.conf
 - /etc/resolv.conf
- dig



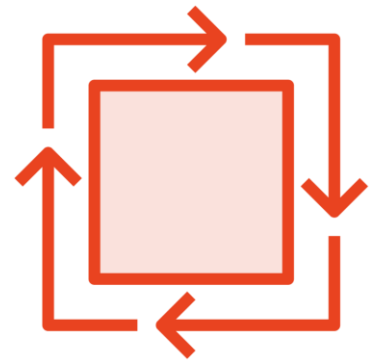
Machine Metadata



hostname: /etc/hostname



pretty name: /etc/machine-info



machine id: /etc/machine-id

```
$ hostnamectl  
$ sudo hostnamectl set-hostname Andrew\'s  
$ sudo hostnamectl set-deployment dev  
$ sudo hostnamectl set-location Chicago
```

Unify Configuration Using Systemd Eco-System

The hostnamectl command can consolidate and display machine information. It can also manage configuring fresh data via the the systemd-hostnamed service

Demo



Configure Hostname:

- hostnamectl
- /etc/hostname
- /etc/machine-
- info / id
- /etc/os-release



Most distributions will check
/etc/hosts before DNS for
hostname resolution. It is not
necessarily the most secure!



```
$ grep '^hosts:' /etc/nsswitch.conf  
$ grep 'ReadEtcHosts' /etc/systemd/resolved.conf
```

Name Service Switch

Name services can be located through database names in the `/etc/nsswitch.conf`. This includes users, as well as hosts. Changing the search order can prevent simple attacks. On Ubuntu, we also have to change the local DNS server not to read the hosts file

Demo



Let's investigate name resolution ordering:

- `/etc/hosts`
- `dns`
- `/etc/nsswitch.conf`
- `getent hosts`



Name Resolution

/etc/hosts

IP full-name aliases

man 5 hosts

DNS

Listens Port 53

systemd-resolved.service



```
$ sudo vim /etc/systemd/resolved.conf  
$ sudo systemctl restart systemd-resolved.service  
$ resolvectl status  
$ ping ubuntu2
```

LLMNR

Local-Link Multicast Name Resolution: If this is enabled on systems, name resolutions can be made simple on local area networks. Multicast requests are sent on the network for hostnames, servers can respond if they match the request. This needs to be enabled on all systems you want names resolved

Demo



DNS Runs Locally on Ubuntu:

- `resolvectl status`
- `/etc/resolv.conf`
- `/etc/systemd/resolved.conf`
- `systemd-resolved.service`
- `dig`



Summary



Hostnames and Name Resolution

- hostnamectl
- hostname
- DNS and Port 53
- /etc/nsswitch.conf
- getent
- resolvectl
- systemd-resolved
- /etc/systemd/resolved.conf



Implementing an NTP Client in Ubuntu Linux

